Tutorial 2: 3D Geometry

Homework: Exercises marked with red stars, with points adding to 10.

Surfaces and curves in space

Space curves

Just like a planar curve, a curve in $\mathbb{R}^3$ can be parametrized with a function that assigns a point in space to a value of a parameter. This point now has three coordinates. Other than that, there is not much difference in the method. There is a big difference though: space curves can not be described with an equation. An equation now contains *three* variables, say x, y and z, and in that case it describes a *surface*. For example, the equation $x + y + z = 1$ describes a plane that contains the points $(1, 0, 0)$, $(0, 1, 0)$ and $(0, 0, 1)$. If you want to use equations to describe a curve, you need *two* equations. Each equation describes a surface, and the curve is simply the intersection of the two surfaces.

Surfaces in space

A surface in $\mathbb{R}^3$ can be defined as the graph of a function of two variables. The **graph of** $h(x, y)$ is defined as the set of points $\big(x, y, h(x, y)\big)$, where (x, y) are points in the domain of f. For example, the graph of the function $h(x, y) = \sqrt{1 - x^2 - y^2}$ is a semi-sphere, namely the part of the unit sphere above the xy-plane.

Observe that for a point (x, y, z) on the graph of f we must have $z = h(x, y)$. The latter is an equation with unknowns x, y and z. In general, you can use any equation in x, y and z to describe a surface. For example, for a point (x, y, z) on the semi-sphere described above, we can derive that $x^2 + y^2 + z^2 = 1$. Note that this equation describes the complete sphere, not just the upper part.

Contours in space

You can regard surfaces in space as the level set of a function of *three* variables. For example, if we define $H(x, y, z) = x^2 + y^2 + z^2$, then the unit sphere can be seen as the set of all points (x, y, z) in space such that $H(x, y, z) = 1$. We call the set of all points with function value C the **level set** at level C. Sometimes they are also called contours, but we like to reserve this term for curves. Level sets in space are usually, but not always, surfaces. Just as with contours, level sets at different levels can not intersect. That's why they often form a onion peel-like structure. See also exercise 2.

EXERCISES

1

You can regard the grain pattern in a wooden plank as a collection of contours. Of which function are they contours?

2 Plot (or draw) several level sets of the function $H(x, y, z) = x^2 + y^2 + z^2$, but only the part of the level set that is below the xy-plane.

Curvature

EXERCISES

H **3** Consider the surface given as the graph of the function $h(x, y) = x^2 + xy$ for all $(x, y) \in \mathbb{R}^2$.

3 (a) Compute the mean curvature H and Gauss curvature K of this surface at the origin $(x, y, z) = (0, 0, 0)$.

1 (b) Is it possible to bend a piece of paper into this surface? Justify your answer.

2 (c) What happens to the curvatures as one moves very far away from the origin?

4 Let's have a look at the surface S given as the graph of the function $h(x, y) = (x^2 - y^2)/2$ over the unit ball $\{(x, y) \mid x^2 + y^2 < 1\}$.

(a) Compute the mean curvature H and Gauss curvature K of this surface at the origin $(x, y, z) = (0, 0, 0)$.

(b) Is it possible to bend a piece of paper into this surface?

(c) Define new coordinates u, v in the XY plane by

$$\begin{cases} x = u + v, \\ y = u - v. \end{cases}$$

What does the equation for the surface S become? Use this new equation to show that S is **double ruled**, that is, it can be constructed out of two families of straight lines.

(d) Is it possible to construct a sliceform of S using polygon slices (that is, with all of their sides straight)? Why or why not?

Projections

EXERCISES

5 Imagine that you are standing atop the middle of a pair of train tracks, which are perfectly straight on completely flat ground, and you are looking along them.

(a) Draw a sketch of what you would see in the real world.

(b) Draw a sketch of what you would see in orthographic projection.

H 6 Assume we have two spheres S_1 and S_2 of radius R_1 and $R_2 = 3R_1$, of different colors, and with their centers on the optical axis at distances d_1 and d_2 from the projection center (pinhole) respectively, with $d_2 \geq d_1$. We observe them with a focal length f_L.

2 (a) Keeping d_1 fixed, how large does d_2 need to be so that we stop seeing S_2 because S_1 occludes it?

2 (b) If we keep the distance between the balls $d_2 - d_1$ fixed but we can move both of them back and forth, at which distance d_1 will both project to the same circle in the image?

Non-orientable surfaces

EXERCISES

7 We will do a couple of experiments around Möbius strips. In each case, you need a long strip of paper and glue or staples to close the ends together. Remember that if we rotate one of the ends of the strip of paper by 180 degrees (half a twist) and glue it to the other end, we end up with a Möbius strip.

(a) What happens if you do a complete twist of 360 degrees, instead of a half one, before gluing the ends together? And if you do one and a half twists?

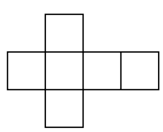
(b) Draw a line through the middle of the strip, in the "long" direction. Twist and end once and glue to obtain a Möbius strip, as usual. Now, cut the strip through the line. What happens?

Polyhedra

Polytopes are geometrical constellations of arbitrary dimension with flat sides. With 'flat' we mean lines and planes. In general, the sides of a polytope are polytopes themselves, but with lower dimension: a cube is a three-dimensional polytope with squares as sides, which themselves are two-dimensional polytopes. And squares have one-dimensional sides (line segments).

A three-dimensional polytope is called a **polyhedron**. The sides of a polyhedron are polygons. The side of a polyhedron is sometimes referred to as a **face**. A cube is a polyhedron. The faces of a cube are squares, which are polygons.

EXERCISES

8 A **net** is a two-dimensional figure that can be folded into a three-dimensional object, e.g.  can be folded into a cube. How many essentially different such cube unfoldings exist?

9 ***The Hypercube***

The analogue in dimension four of the cube (or square) is the so-called **hypercube**[1]. It consists of 16 vertices. Each vertex is connected to 4 other vertices by an edge of unit length. Compare this to a 3-dimensional cube.

(a) In coordinates the vertices are (a_1, a_2, a_3, a_4) with $a_i = 0$ or $a_i = 1$. Describe a condition for two vertices to be connected by an edge.

(b) Deduce how many edges a hypercube has.

[1] The hypercube is also known as the **tesseract**.

Hints

1 How are trees made?

3 (a) Remember the formulas for surfaces given as graphs from the lecture slides. What is the tangent plane at the origin?

(b) The question is not about the shape we would have to cut out of the paper, but about the surface itself. You can use the answer to the previous question.

(c) Computing them is harder than in the first question, because the tangent plane is not always the same.

4 (a) Like in the previous exercise, curvature plays a role here.

(b) Remember that if in the parametrization of a surface we fix one variable to some value, then we are drawing a curve.

(c) With polygon slices, we mean that the cuts we make in a piece of cardboard are straight.

5 (a) No hints for this one.

(b) Are you looking straight at the horizon, or down at the tracks?

6 You can think of coordinate systems (x, y, z) in space with the spheres being positioned in the z axis, and (X, Y) in the image plane (which is perpendicular to the z axis).

8 The squares in the net must be connected. Rotating a net does not constitute for a different net, nor does shifting or reflecting.

9 (a) First try to find a similar condition for the vertices of a three-dimensional cube.